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COMMENTS:

Introduction

Hierarchical Task Analysis (HTA) is a fundamental, structured approach within Human-Computer Interaction (HCI) that serves to break down a user's primary objective into smaller, more manageable subtasks. This systematic decomposition starting from the overall objective and proceeding through sub-goals, operations, and detailed actions provides a clear, step-by-step understanding of how users naturally perform their activities. By mapping this logical flow, HTA helps designers identify user behaviors, critical decision points, potential areas for error, and key areas where system support or improvements are most needed. The insights gained from HTA are essential, as they clarify the logical flow of tasks, expose inefficiencies, and directly inform the creation of precise design requirements, ensuring the system is built around verified user needs rather than mere assumptions. These requirements typically cover functional specifications (what the system must do), usability needs (how intuitive and easy the system should be), and contextual considerations (where and by whom the system will be used).

This analysis applies the HTA methodology to the **Recircle app**, a system designed to encourage and facilitate user participation in recycling. Recircle offers crucial features, including the ability for users to schedule a convenient **door-to-door collection pickup** (Task 1), a feature that is a critical functional requirement, particularly for users with large volumes of waste. It also provides a **location service** (Task 2) to help users find the nearest suitable drop-off point and verify what materials are accepted there. Furthermore, to drive engagement and motivation, the app incorporates a **Rewards Points/Cash Redemption System** (Task 3), allowing users to convert accumulated recycling points into cash value, vouchers, or gift items.

The HTA in this report focuses on analyzing the user experience of key target demographics such as the College Student, Small Business Owner, and Waste Management Worker as they interact with these core features. By dissecting these tasks, the HTA reveals specific friction points: for example, the difficulty users face when estimating waste quantity for pickup (Task 1) or the usability challenge of confirming accepted materials at a drop-off point (Task 2). The resulting findings from the HTA directly lead to the derivation of essential **design requirements** what to keep, what to redesign, and what to add—ensuring Recircle's evolution towards a more efficient, user-friendly, and purpose-driven platform

Gathering Requirements - Task Analysis

Derivation of HTA:

Target User : College's Students

Link 1 : <https://youtu.be/bmLnAwzVZsE>

Link 2 : <https://youtu.be/e6GoVSXtrU8>

Link 3 : https://youtu.be/QUL8-ZN_Ht0

Finding from the HTAs :

Task 1:

The goal of this task is to successfully submit a request for a door-to-door recycling pickup. The user, a Busy Senior (Aina), begins with the external preparation phase, which involves collecting and sorting the items, followed by the crucial and most challenging sub-task: estimating or weighing the quantity of each segregated material. Once the items are prepared, the user moves through the sequential digital process: starting the request in the app, selecting the appropriate waste categories, and inputting the estimated quantities. The user then confirms the automatically populated pick-up location (their dorm address) and selects the desired time slot. The final operation is submitting the request after a quick review. The overall Plan for this task is linear: complete preparation, then execute the four main in-app operations sequentially (Start Request, Specify Items, Confirm Details, Submit Request).

Task 2 :

The goal for the Busy Senior in this task is to locate the nearest suitable drop-off point and verify if a specific material (like used cooking oil) is accepted at that location. The user's primary plan involves searching for the location feature and immediately tapping the nearest hotspot on the map to review its details. This reflects the user's desire for speed and convenience. The most critical decision point—the contingency plan—occurs after viewing the location details. The user ideally expects to see the list of accepted materials on the location's dedicated detail page. If that information is absent, the user must abandon the location view, navigate back to the main menu, and perform a separate search within the app's overall recycling categories to confirm acceptance. This deviation from the ideal path highlights a key area for design improvement.

Task 3 :

The goal of this task is to convert accumulated reward points into cash value. The HTA for this task starts with a pre-condition, which is the overall Plan: the user must first check the point balance upon entering the Rewards section. Only if the points are sufficient for a meaningful cash-out amount will the user proceed with the core sequence. This sequence involves initiating the cash redemption option, then securely inputting the required bank details (Name, Bank, Account Number, and the point amount). After reviewing the redemption summary, the user confirms the transaction and then checks the history to verify the request is pending. If the initial point balance is deemed too low to proceed, the user terminates the task, saving the input effort for a later, more valuable redemption.

Target user : Small Business Owner

Link 1 : https://youtu.be/d44FylsjB4g?si=Y2p0QO23YQjb_uIs

Link 2 : <https://youtu.be/ykUjTagoZfs?si=CC2c3IQDSGZ7gQC7>

Link 3 : <https://youtu.be/kZyZKWUEhVk?si=0cl0EP4DaBm8c2W8>

Finding from the HTAs :

Task 1 :

The goal of this task is for a Small Business Owner (Zura) to successfully schedule a pickup for used cooking oil through the Recircle app. The task begins with the external preparation phase, where Zura filters the used cooking oil, transfers it into sealed containers, and estimates the quantity based on container size and weight. This estimation step is the most challenging because used oil containers are heavy, and Zura needs to calculate the volume manually.

Once preparation is complete, Zura moves to the digital phase. She begins by starting the pickup request in the Recircle app. However, Zura faces a usability barrier such as the small plus button under the recycling category section is difficult for her to notice, causing confusion about how to add the waste item. After locating the button, Zura continues to the item specification stage, where she selects “Used Cooking Oil” as the waste category. Another challenge appears here, as the app does not clearly indicate whether the quantity should be entered in kilograms or liters, leaving Zura uncertain about which unit the system expects. After making her best estimate, she proceeds to confirm the business address automatically populated by the app and selects a pickup time slot that aligns with his operating hours.

Finally, Zura reviews the request details and submits the pickup request. The overall plan for this task remains linear, complete preparation, then execute the four in-app operations sequentially (Start Request, Specify Items, Confirm Details, Submit Request), with the added challenge of unclear UI elements affecting Zura’s experience.

Task 2 :

The goal of this task is for Zura, a Small Business Owner, to locate the nearest suitable drop-off point and verify whether her used cooking oil is accepted at that location. To achieve this, she opens the Recircle app and navigates to the location feature, scanning the map for the closest hotspot that could serve as a recycling point. She taps on the nearest location to review its details, intending to quickly determine if it can accommodate her specific type of waste. However, Zura immediately notices a usability challenge: the list of accepted items is displayed in very small text, making it hard to read, and the items are not organized according to recycling categories.

This forces her to spend extra time trying to locate her material within the list, which can be frustrating and increases the risk of misinterpretation. Because the information is unclear and difficult to confirm, Zura is left uncertain whether her used cooking oil is actually accepted at that location. Faced with this ambiguity, she ultimately assumes that her material is accepted and proceeds without verification. This highlights a significant area for improvement in the app’s interface, as clear, well-organized, and readable information about accepted materials would help users like Zura make informed decisions and reduce the likelihood of errors or unnecessary assumptions.

Task 3 :

In this task, Zura wants to redeem rewards using the recycling points she has earned. The Rewards feature encourages users by offering incentives like vouchers, discounts, or gift items for their recycling activities. She starts by going to the Rewards section in the app, which can usually be found in the bottom navigation bar or main menu. When she taps the Rewards icon, the app shows her current points balance so she can see how many points she has earned. The app then displays rewards she is eligible to redeem. Each reward shows important information, such as the required points, type of reward, validity period, and a brief description. This helps Zura compare the options and decide which reward she wants. However, she notices a usability issue: some expired rewards still appear in the list, which could cause confusion and make it harder for her to identify valid options.

After selecting a reward, the app shows a confirmation page summarizing the reward details and the points that will be used. She taps the Redeem button to confirm, and the system processes her request and updates her points balance. Once the redemption is successful, the app provides a confirmation message along with any necessary code, voucher, or digital receipt. The redeemed reward is also saved in the My Rewards or Reward History section for future reference. This feature allows Zura to enjoy the benefits of her recycling efforts and motivates her to continue participating in the program, though improving the display to hide expired rewards would enhance the user experience.

Target User : Waste Management workers

Link 1 : https://youtu.be/n8-n2_xHIk0

Link 2 : <https://youtu.be/XKEfGdZeLbU>

Link 3 : <https://youtu.be/yCIOemIUXTM>

Finding from the HTAs :

Task 1:

In this task, the focus is on how Mr. Fakhrul initiates and completes a recycling request using the mobile application. The process begins when he taps on the Recycle icon located on the bottom navigation bar. This icon acts as the main entry point for users who wish to schedule a pickup for their recyclable items.

Once the recycle option is selected, the system redirects Mr. Fakhrul to the Recycling Request Form. At this stage, he is required to provide all the necessary information to ensure an accurate and convenient pickup. One of the key details he must enter is the pickup location. This allows the collection team to know exactly where the recyclables will be retrieved. The location can be entered manually or selected using the app's integrated map feature, depending on the system's design.

After confirming the pickup location, Mr. Fakhrul proceeds to complete the remaining required fields, which may include the type of recyclable materials, estimated quantity, preferred pickup date and time, and any additional instructions for the collector. The system verifies the information and allows him to review the details before final submission.

By clicking Submit, the app successfully records his recycling request. The system then generates a confirmation message to inform Mr. Fakhrul that his request has been received and is being processed. This streamlined interaction ensures users can create a recycling request efficiently and with minimal steps.

Task 2:

For this task, Mr. Fakhrul intends to locate the nearest recycling centre using the application. This feature is designed to help users quickly identify nearby facilities along with all relevant details before they decide to visit.

The process begins when Mr. Fakhrul navigates to the Recycle Centre section within the app. Upon selecting this feature, the system automatically detects his current location or prompts him to choose a preferred area. Based on this information, the app displays a list or map view of the nearest recycling centres available around him.

When Mr. Fakhrul selects a specific centre from the list, the application provides a detailed information page. This page includes key details such as full address, contact information, operating hours, types of material accepted

Task 3:

In this task, Mr. Fakhrul wishes to redeem rewards that he has earned through his accumulated recycling points. This feature is designed to motivate users by offering incentives, such as vouchers, discounts, or gift items, in exchange for consistent recycling activities.

The process begins when Mr. Fakhrul accesses the Rewards section of the application. This feature can typically be found on the bottom navigation bar or within the main menu. Once he taps on the Rewards icon, the system retrieves and displays his current points balance, allowing him to see how many points he has earned so far.

Based on his available points, the system displays a variety of rewards that are eligible for redemption. Each reward item includes essential details such as the required number of points, reward type, validity period, and a short description of the benefits. This helps Mr. Fakhrul compare different reward options and choose the one that suits him best.

When he selects a reward he wants to redeem, the app shows a confirmation page summarizing the reward details and the points that will be deducted. After reviewing, Mr. Fakhrul proceeds by tapping the Redeem button. The system then processes the request and updates his points balance accordingly.

Upon successful redemption, the system generates a confirmation message, along with any redeemable code, voucher, or digital receipt that Mr. Fakhrul needs to use the reward. The redeemed reward is also stored in the My Rewards or Reward History section for future reference.

This feature allows Mr. Fakhrul to gain tangible benefits from his recycling efforts, encouraging continued participation and engagement with the recycling program.

Design Requirement

What to Keep (Successful and Core Functions)

The core value proposition and functionality of the Recircle application must be preserved. Specifically, the system must keep the door-to-door collection scheduling (Task 1) and the Reward Points/Cash Redemption System (Task 3). The door-to-door service is a critical functional requirement (FR4) for college students, particularly for the Busy Senior clearing out large volumes of waste during move-out periods, offering unparalleled convenience that reduces the "cost" of recycling. Similarly, the points system and cash conversion (FR5) are highly effective motivational drivers for this demographic, transforming recycling into a valuable, tangible activity. The current task structure for initiating a request and redeeming cash, while needing minor simplification, is fundamentally sound and successfully meets the users' primary goals.

What to Redesign and Add (Addressing Pain Points)

The HTAs likely revealed key inefficiencies, particularly around knowledge and data entry, necessitating new functional requirements. The most significant redesign involves the recycling classification process. To address the lack of sorting knowledge observed in the Conscientious Freshman, a new feature for Simplified Material Identification (FR1) is required. This should include an Image Recognition tool or a Barcode Scanner that allows users to quickly verify an item's recyclability before starting a pickup request, simplifying the subtasks involved in categorization. Furthermore, the HTA for Task 1 likely showed friction when the user must input an item's weight; therefore, the app must incorporate an Estimated Weight Guidance (FR3) feature. This feature should use visual analogies (e.g., "This amount of paper is ~5kg") to replace arbitrary number entry, making the task flow more efficient for the Busy Senior and reducing the chance of user abandonment.

Usability and Security Requirements

Finally, the non-functional requirements must be enhanced to fit the college context. Usability (NFR1) is paramount: the app's navigation requires consolidation, as demonstrated by the user's struggle to confirm accepted materials and find a drop-off location simultaneously (Task 2). The system should integrate all on-campus "Micro-Drop" Location Services (FR2) and their accepted materials list into a single, intuitive map interface, optimizing the task flow for quick drop-offs. Crucially, the app must satisfy the Security (NFR2) concerns of the Tech-Wary Commuter during the cash-out process. This requires adding clear Trust and Security Assurances on the bank detail entry screen and providing prompt, transparent communication about the fund transfer status to ensure the user feels confident and safe completing the redemption task.

Gathering Requirements - User Analysis

Proposed Task

2.1 Assessment of Proposed ReCircle Features against Identified User Points

The three proposed tasks Real-Time Camera Scanning, Guided Multi-Component Sorting and the Recycling Scorecard are highly effective in addressing core issues in mitigating critical user pain points identified during preliminary user feedback and operational reviews.

2.1.1 Core Feature Effectiveness (Quality Control and User Engagement)

The three proposed tasks demonstrate a strong focus on resolving issues related to material quality, contamination and sustained user motivation:

Identify Waste Category using Real-Time Camera Scan , this feature is designed to provide immediate feedback on the disposal category (recycling, landfill, compost and special). Mitigation of rejection Quality by providing instant authoritative categorization, this system significantly reduces the instance of incorrect items being included in the recycling submission. This directly addresses the request that is difficult to approve due to the low quantity of actual recyclable material. Improved material purity reduces the probability of rejection by collection partners. Addressing Low Volume while this feature does not increase the physical quantity of materials a user has, it ensures that 100% of the materials submitted for recycling is correctly classified increasing the material's market value and making smaller collections more economically viable for the collection partners.

Sort Multi-Component Items with Guided Instructions This function breaks down complex products like layered food packaging and liquid containers into their individual components, instructing the user on the correct disposal for each particular item. Enhancing purity this critical contamination mitigation strategy. Contamination often occurs when components are separated like liquid remaining in a container. By guiding separation this feature elevates the purity of the submitted recyclable material directly addressing the difficulty in securing approval for collection requests by minimizing post-collection sorting effort required by partners.

View and Track Recycling Scorecard this task involves presenting users with metrics on their recycling performance, accuracy and weekly improvement supported by a gamified point system. Driving sustained usage this feature is essential for addressing the strategic challenge will this app make users want to use it more. The gamified scorecard leverages behavioral economics to convert a chore into a rewarding activity. Tracking progress, earning points and viewing improvements serve as intrinsic motivation, drastically improving user retention and boosting overall app usage.

Persona

User Group 1 : Wirda Alissa (19 Years Old University Student)

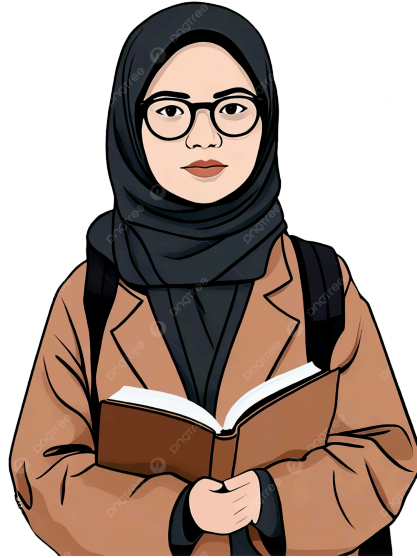


Figure 1: Wirda Alissa (University Student)

Wirda is a highly motivated first year student enrolled in the Computer Science program, and she lives in the university dorms. Although she has a strong motivation to be sustainable, she really cares about the environment because of what she has learned from university programs and global goal campaigns she constantly struggles to recycle correctly. The main barrier is the recycling system in her building. There are only a few communal bins scattered around the dorm, and their labels are often confusing or sometimes even missing, making it hard to know what goes where. Because Wirda is always rushing between her classes and homework, she needs technology that is fast, simple, and requires almost no thinking. She prefers mobile apps that she can use quickly on her phone, which is a tool she relies on all day long.

Wirda's motivation to download and use the ReSort app is not just about helping the environment; it is also personal. She wants to use it to finally establish positive, lasting habits around waste disposal. She often feels a wave of guilt and frustration when she throws an item into the general trash, only to realize later that it was recyclable. This happens mostly because she is trying to save time when running out the door. She believes ReSort will solve this by letting her use the camera to instantly scan the item and get a clear, quick answer, which is much better than stopping to search online every time. Her core goals are to eliminate those recycling mistakes, see her weekly score go up, and achieve a state of real confidence where she never has to doubt her disposal decisions again.

User Group 2 : Kak Mariam (40 Years Old Canteen worker)



Figure 2: Kak Mariam (Canteen Worker)

Kak Mariam is a 40 years old canteen worker who handles food preparation, serving and cleaning every day. She deals with a lot of packaging for instance plastic cups, straws and food containers. During peak hours, she usually will throw everything together because she does not know how to separate the different materials properly. She wants to recycle but finds it pretty confusing when one item contains many parts and searching it online is very time consuming. She needs something fast, clear and easy to follow.

Her biggest challenge is figuring out how to sort mixed material items correctly especially when there are cups with plastic lids, foam and food packaging with multiple layers. This makes her interested in the ReSort app which offers a feature called Guided Multi Component Sorting. This feature can break down each item and show her which part needs to be separated which does not. With this feature, she hopes the app can make sorting faster so she can keep the canteen clean without slowing down her work.

User Group 3 : Muhd Hafiz (25 years old Engineer)



Figure 3: Hafiz (Engineer)

Hafiz is a 25 years old engineer who works in an industrial environment and handles different materials such as plastic, cardboard and packaging from equipment. He tries to recycle properly, but most of the app does not track how well people are sorting their waste. For Hafiz particularly, he likes a feature where it can show progress, achievements and statistics. He is drawn to technology that helps him improve his habits. Due to his busy schedule, he prefers tools that help him to stay consistent without needing some extra effort.

He is especially motivated by the Resort Recycling Scoreboard feature as it allows him to track recycling accuracy and improve over time. With number points and weekly progress can help Hafiz maintain good habits. With the scoreboard, he can see if he is getting better and even compare his performance with his coworkers as a fun challenge. The motivation and clear feedback make him more committed to recycling not only at work but also at home.

Scenarios

Task 1 : Identify Waste Category Using Real-Time Camera Scan

Wirda is a busy first-year college student, and this morning is no different. She just finished eating her instant noodles, and she is running late for her 8 AM lab class across campus. She only has a couple of minutes to clean up and get out the door. She looks at the leftover packaging and immediately feels confused. She has a paper cup, a clear plastic lid, and a small seasoning packet. She knows they probably don't all go in the same place, but she can't remember which bin to use for each one.

In the past, Wirda would waste precious time searching on her phone, typing in something like "is paper cup recyclable?" and getting unclear answers. Today, she decided to use ReSort. She quickly opens the app and points her phone camera at the three items. The real-time scan feature works right away. The app gives her simple, clear instructions: it tells her the paper cup must go in the landfill bin because of the thin wax coating inside. It directs the plastic lid to the recycling bin. Finally, it tells her the seasoning packet also goes to the landfill. Wirda manages to sort and throw away all three items in less than 10 seconds. She feels a huge sense of relief because the quick camera check helped her avoid making mistakes, giving her confidence that she is doing the right thing, even when she is rushing.

Task 2 : Guided Multi-Component Sorting

During the after school lunch cleanup, Kak Mariam notices a pile of used drink cups, plastic lids and food containers left on the counter. She wants to recycle them but she gets confused whenever an item has several parts especially during rush hours. Instead of guessing, she opens the ReSort app and chooses the Guided Multi-Component Sorting feature. She scans a drink cup with a sealing film and straw in hope that the app will clearly show her how to separate everything correctly.

Within a few seconds, ReSort breaks the item down into different components and tells her exactly what to remove and which part to rinse. These simple steps make it easy for her to sort the items even when the canteen is busy. Kak Mariam quickly separates the cup, sealing film and straw without worrying if she is doing it wrong. After using the app a few times, she feels more confident and realizes that sorting waste properly does not slow her down anymore.

Task 3: Recycling Scorecard

At the end of his shift, Hafiz is curious about how well he has been recycling throughout the week. Since he handles a lot of packaging in the factory, he uses ReSort regularly to scan items and sort them properly. He opens the Recycling Scorecard feature to check his accuracy, points and weekly progress. The scoreboard shows how consistent he has been and whether his recycling habits are improving compared to the previous week.

When Hafiz sees his accuracy increase, he feels more motivated to keep improving. The app shows his streak and gives him badges for staying consistent which makes the process feel more fun and rewarding. He decides to challenge himself to get an even higher score next week. The scorecard helps him stay aware of his habits and makes him more responsible with waste sorting at work and even at home.